

AIRIAS: Revolutionizing Long-term Ocular Management For Stevens-Johnson syndrome with AI-Powered Real-time Image Analysis

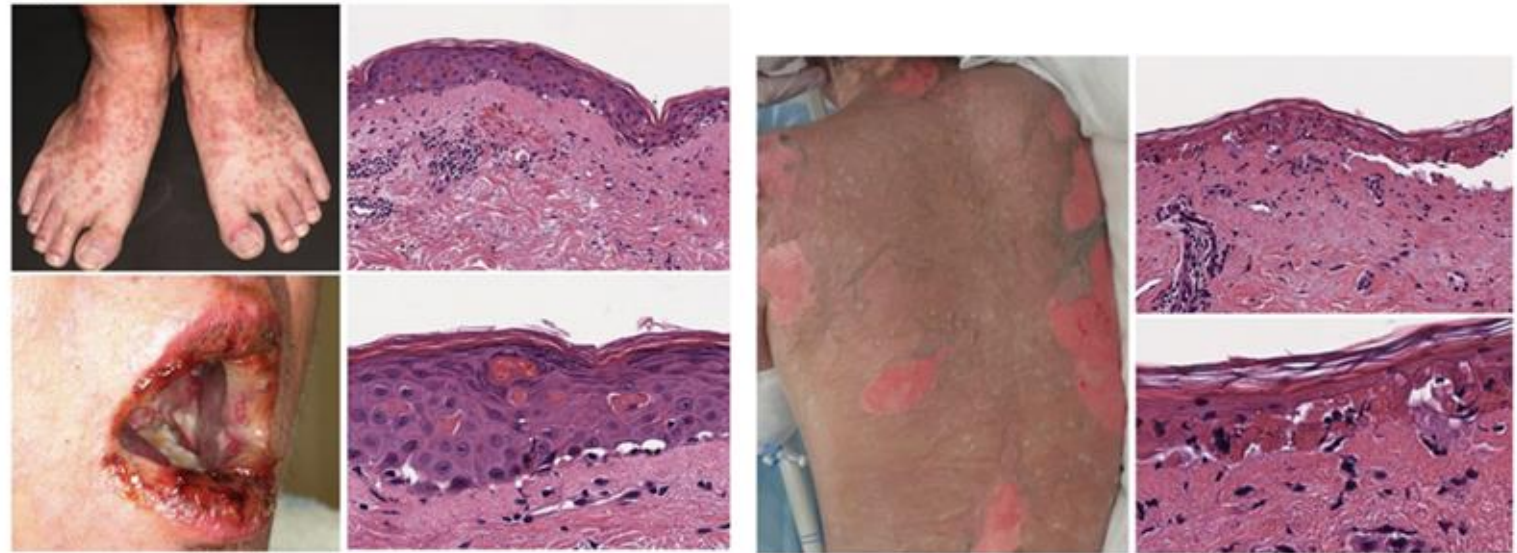
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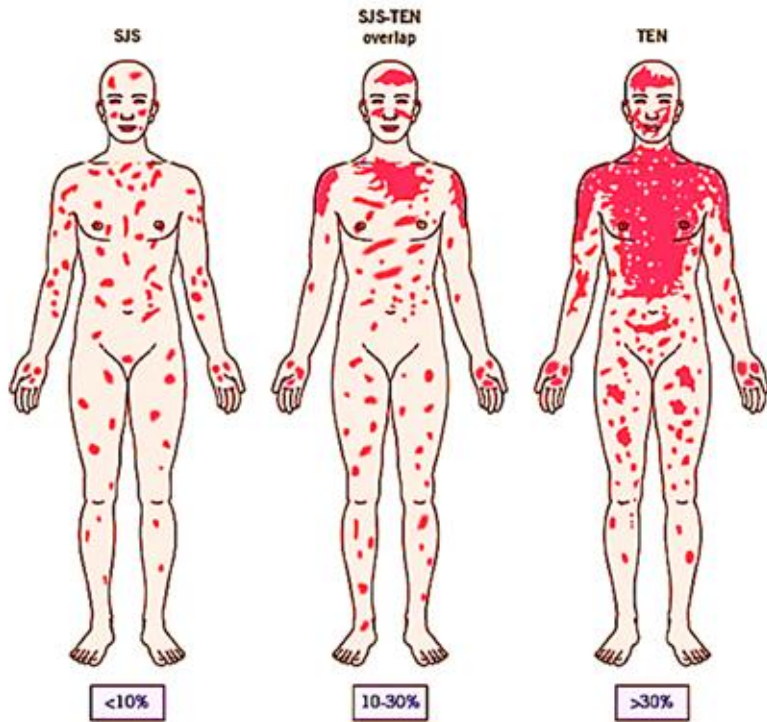
The Challenge of Stevens-Johnson syndrome (SJS) Diagnosis: A Patient's Journey

Delayed Diagnosis: The optimization of care for chronic phase multi-organ involvement after acute phase care at ICU is overlooked.

Lack of Expertise: PCPs and optometrists lack the experience to accurately assess multi-organ complications of SJS and accordingly refer the patients to tertiary medical centers for further tx.

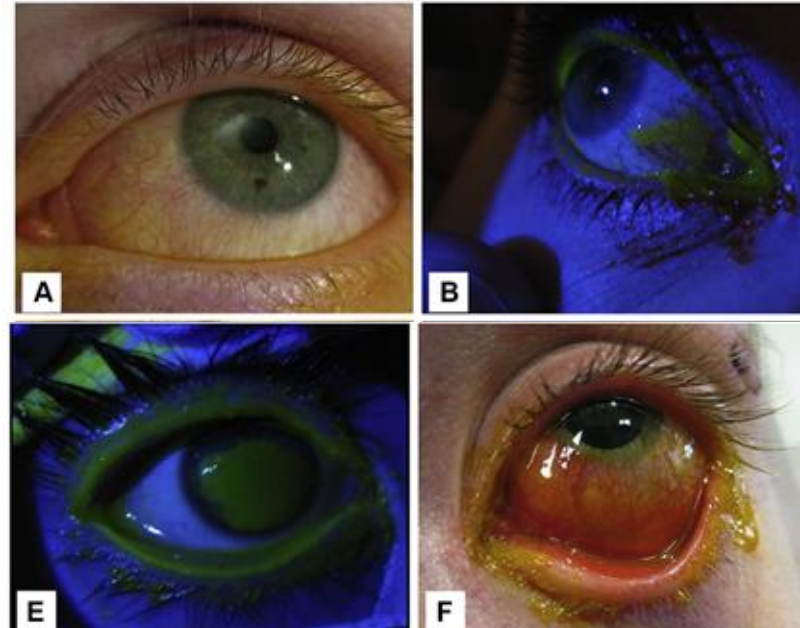
Time Sensitivity: Early diagnosis and intervention are critical in preventing long-term sequelae.





SJS/TEN as a Rare Disease

- **Rare, life-threatening immune reaction** triggered by medications or infections. Affects skin and mucous membranes, including the **eyes**, oral mucosa, and gastrointestinal tract.
- **Incidence:** Affects **1-2 cases per million people annually**.
- **Mortality Rates:** **SJS:** ~5-10% fatality rate. **TEN:** Up to **30-50%** fatality rate.
- **Long-Term sequelae:** **50-80%** of survivors develop **severe eye complications**.
- **Phase of involvement:** Rash, blistering, eye inflammation in acute phase, scarring, dry eye, and **permanent vision loss** in chronic phase.



Mild ocular SJS

Severe ocular SJS

What we bring to the table

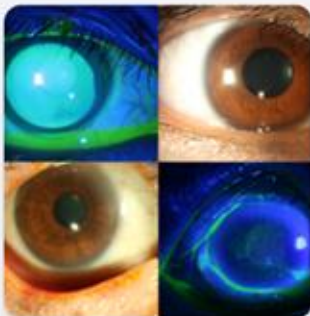
- **Product:** **AI**-Powered **R**real-time **I**Image **A**Analysis**S**
- **Solution:** CNN-powered real-time image Analysis
- **Input:** PCP or optometrist-uploaded eye images (visible light photos or fluorescent staining photos)
- **Output:** Automated assessment of acute phase and chronic phase ocular involvement of SJS





SJS Diagnosis AI Assistant

This project focuses on utilizing external eye photographs from both acute and chronic stages of Stevens-Johnson syndrome (SJS), which is a severe immune-related adverse event often triggered by medications or infections. The primary objective is to develop a machine learning algorithm to identify the severity of corneal involvement in SJS accurately.



Start Diagnosis

AI Diagnostic Assistant

How long has the patient been experiencing symptoms related to Stevens-Johnson syndrome? (e.g., days, weeks, months)

4 weeks

What type of image will you be uploading? (regular or fluorescent)

Regula

Type your answer here...



Upload SJS Image Files



image (1) (1).jfif

0.54 MB

Upload complete!

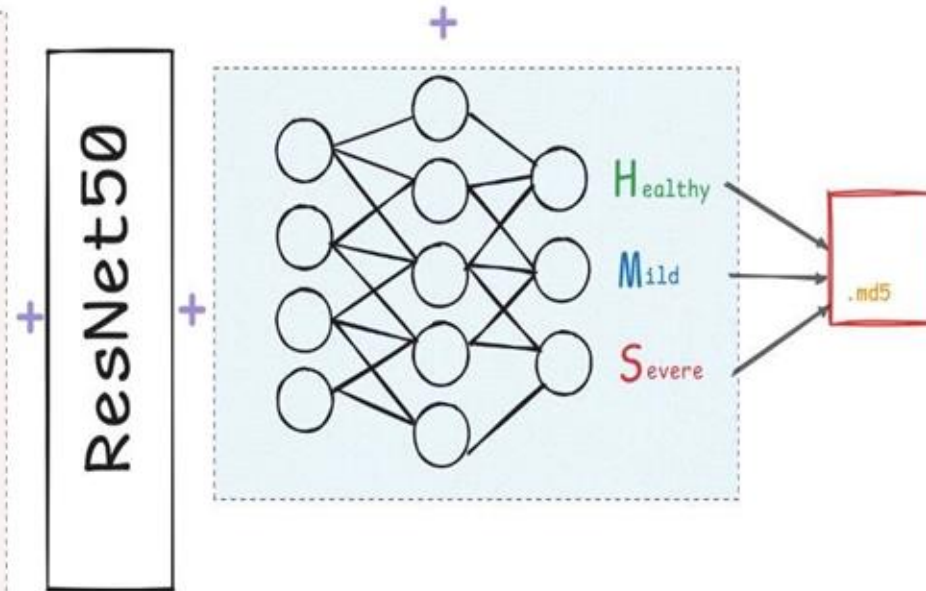
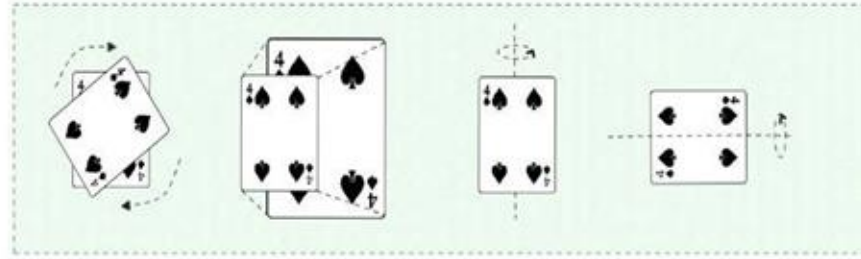
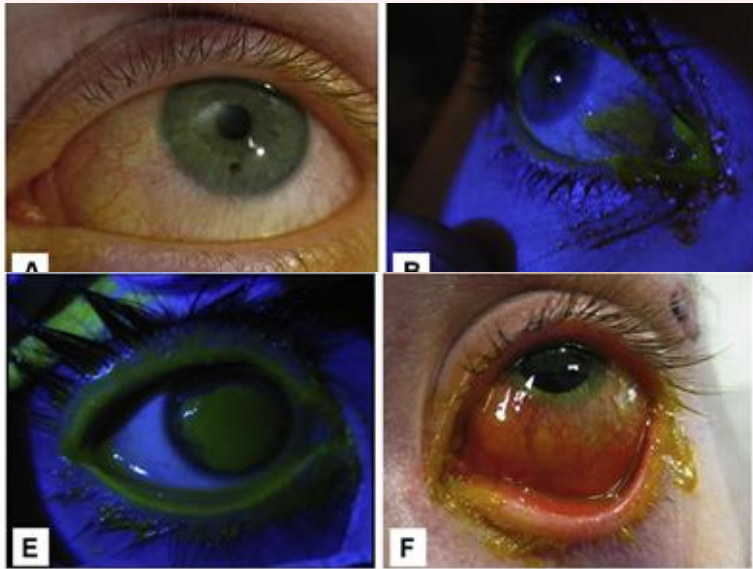
CLEAR

UPLOADED

Supported formats: JPG, PNG, TIFF • Max file size: 10MB

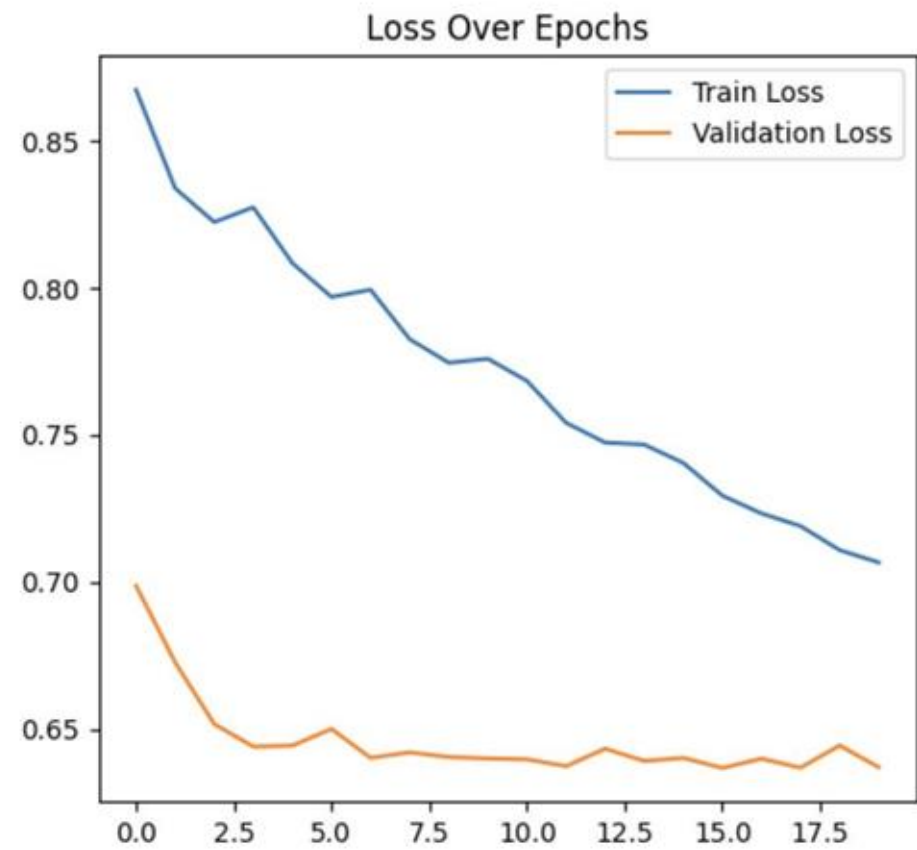
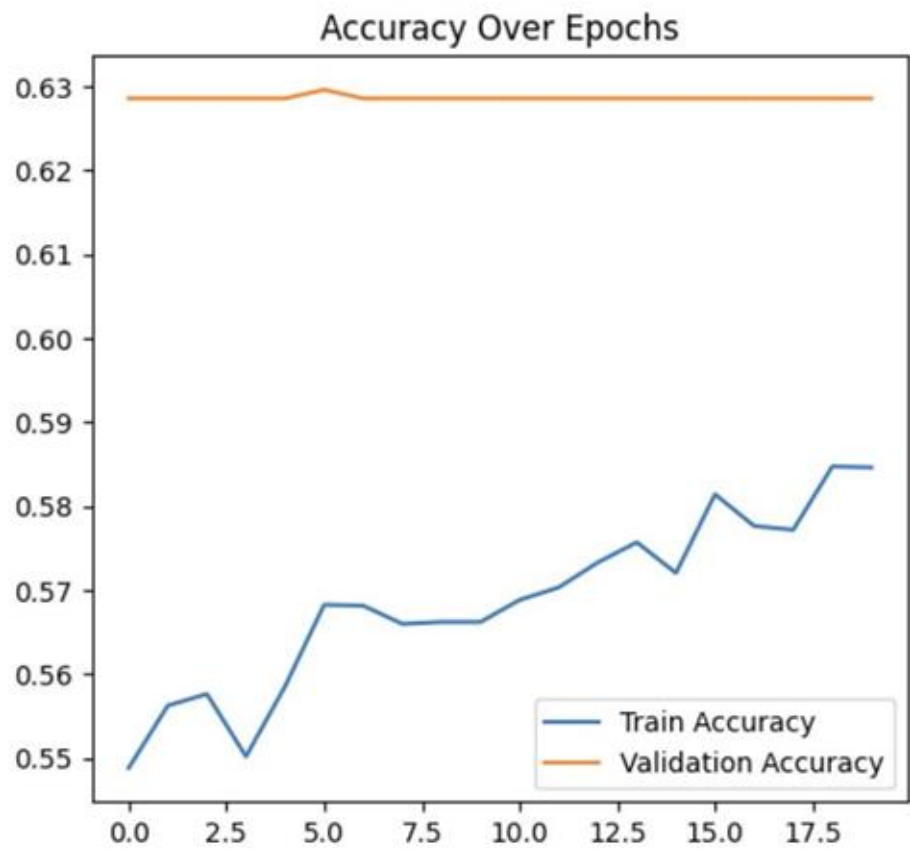


CNN POWERED REAL-TIME IMAGE ANALYSIS



- Multi-center registry of patients with SJS undergoing regular ophthalmic consultations (n= 2k patients).
- In collaboration with ongoing studies at Mass General Hospital.
- Ocular images include visible light photos and fluorescent staining photos at acute and chronic phase.
- Photos were classified into three severity grading (severe SJS, mild SJS, and no SJS ocular involvement).

CNN models achieved an AUROC of 78% in assessing ocular involvement in SJS, including acute or chronic phase, severity and grading of the ocular involvement, unbiased by types of uploaded images!!



Challenges, Discussion & Future Direction

- The model may benefit PCPs or optometrists serving as first-line care providers for patients and can also be used by patients for self-monitoring.
- Further efforts needed to improve training accuracy.
- Collaborations including more healthcare organizations are needed to establish more inclusive patient registry for SJS.
- Collaborations including data input from healthcare professionals from more specialists are needed to facilitate multi-disciplinary symptom management.

